

COMP.2560: System Programming Assignment #2: File I/O (cont.)

ASSIGNMENT 2: Overview

The purpose of this assignment is to evaluate your comprehension on standard and system file I/O calls.

There are five (5) questions in this assignment, described below.

ASSIGNMENT 2 – Question 1

Study the Linux/Unix command “cat” to understand its basic functionality. Then study the attached source code “mycat.c” which implements the “cat” command using standard I/O functions. One can see that this program used “getc (. .)” function to read the content from a file.

Rewrite the attached “mycat.c” program so that it now uses the “fgetc (. .)” function to read the content from a file.

ASSIGNMENT 2 – Question 2

Still on the “mycat.c” file.

This time, rewrite the posted “mycat.c” program so that it now uses the “fread (. .)” function to read the content from a file.

ASSIGNMENT 2 – Question 3

Still on the “mycat.c” file.

This time, rewrite the posted “mycat.c” program using the system I/O functions (open(), close(), read(), write(), etc.).

ASSIGNMENT 2 – Question 4

Using only standard I/O functions (like fopen(), fseek(), ftell(), fgetc(), etc.), write a C program to reverse the order of the lines in a text file. That is, the first line becomes the last, the second line becomes the one before the last one, and so on.

Note: your program should have two arguments, the input file name and the output file name.

Call model: `reverseFile inputFileName outputFileName`

Example:

Input file:

Hello
I am happy to see you

Output file:

I am happy to see you
Hello

ASSIGNMENT 2 – Question 5

Use standard I/O functions to write a C program to combine multiple text files into a single file and display the content of the combined file. Assume we call this program “append”, the way to use it is as follows:

```
shell prompt> ./append file1 file2 file3 combinedfile
```

In the example above, `file1` to `file3` are existing text files, and `combinedfile` is a newly created file whose contents are the concatenation of `file1` followed by `file2` followed by `file3`.

Your program should display the contents of the `combinedfile` in the terminal window before it terminates. Your program should work on any number of input files. The example usage given above shows 3 input files (`file1`, `file2`, and `file3`), and your program should also work, for example, say, 30 inputs files (`file1`, . . . , `file30`) as well.

ASSIGNMENT 2 – Wrap-up and Submission

- For all questions, you need to submit your source code. Submit the source code for each question and include any accompanying files (if applicable) needed for that question
Please make sure all your code works properly on CS Linux server.
 - Submit the link to the video you will create.
 - Please properly name all your files so that markers can easily identify which file is for which question.
See the *Assignment Submission Specification document* for naming files in your submissions for this semester.
 - Record one video with audio showing your source code, you compiling and running each program and explaining how each of the underlined items in each question’s description is realized in your code.
For the convenience of marking your video submission, please add chapters to your video so that markers can easily jump to the start of each question in your video. If you do not know how to add chapters to your video, please watch this short video to learn how:
 - <https://www.youtube.com/watch?v=d5uSWaCPLj4>
- If you are using Microsoft Streams, this link shows how to add chapters:
- <https://support.microsoft.com/en-us/office/using-manual-chapters-with-videos-on-stream-on-sharepoint-8bbf61eb-c00b-42b5-a514-cce2e45eb6ea>
- Please note the late assignment submission policy stated in the course syllabus.